

ASHRITHA G
Software Development Engineer | ashrithag1126@gmail.com

SUMMARY

Software Development Engineer with 3+ years of experience designing and scaling distributed systems, microservices, and full-stack web applications. Skilled in building high-availability APIs and event-driven architectures using Java, Python, and Node.js across AWS and GCP. Proven track record in optimizing performance, automating CI/CD pipelines, and delivering production-grade software with 99.9% uptime. Passionate about system design, observability, and creating reliable, user-centric solutions that bridge engineering, data, and AI technologies.

SKILLS

Programming Language: Java, Python, C++, JavaScript, SQL, C, R

Backend Frameworks: Spring Boot, Spring Security, Spring Data JPA, Hibernate, Node.js, FastAPI

Frontend Frameworks: ReactJS, Redux, HTML5, CSS3, GraphQL

Data & Machine Learning: Pandas, NumPy, Scikit-Learn, PyTorch, TensorFlow, Matplotlib, Seaborn, Data Wrangling, Feature Engineering, Model Evaluation, MLOps

Cloud Platforms: AWS (EC2, Lambda, S3, CloudWatch), GCP, Microsoft Azure

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis, Oracle

DevOps & CI/CD: Docker, Kubernetes, Jenkins, GitHub Actions, Gradle, Maven, CI/CD Pipelines

Distributed Systems & Messaging: Apache Kafka, RabbitMQ, Event-Driven Architecture

Monitoring & Observability: CloudWatch, LogRocket, Prometheus

Collaboration & Agile Tools: JIRA, Confluence, Postman, Agile Development

Software Design: System Design, API Architecture, Microservices Design

WORK EXPERIENCE

Software Engineer Intern | *Outlier AI, USA*

May 2025 - Present

- Developed and optimized interactive web applications using ReactJS, Node.js, and Python, reducing page load times by 20% and improving user workflows.
- Engineered real-time, fault-tolerant data pipelines using Firestore and Redis, enabling sub-100ms query access under 10,000+ concurrent sessions.
- Designed and deployed fault-tolerant microservices on GCP, built CI/CD pipelines with GitHub Actions, and implemented observability pipelines, cutting incident recovery time by 30%.
- Collaborated with product managers and backend teams to refactor API contracts, cutting integration bugs by 35% and aligning features with user needs.
- Integrated application-level caching and load balancing strategies across GCP microservices, improving throughput by 45% during peak traffic.
- Partnered with the DevOps team to automate deployment rollbacks and environment provisioning, reducing deployment errors by 25% and improving release velocity.

ML & Data Researcher | *UMass Boston, USA*

Nov 2023 – May 2025

- Built a cloud-integrated data engineering pipeline in Python to automate ingestion, cleaning, and processing of Low-Income Housing Tax Credit (LIHTC) datasets, ensuring compliance with research data governance standards.
- Optimized data workflows for high-volume public datasets (LIHTC, HMDA) using SQL and Pandas, cutting processing time by 40% and improving reproducibility.
- Partnered with academic stakeholders and machine learning teams to design features and prepare datasets for predictive modeling, boosting accuracy in identifying underfunded housing areas by 30%.
- Applied data validation and anomaly detection techniques to maintain quality and integrity across all processed datasets.
- Developed and maintained analytical reports and visualizations to monitor data pipeline performance and support decision-making for affordable housing policy.

Associate Software Developer | *Wipro, India*

Oct 2022 - Aug 2023

- Built scalable microservices in Java and Spring Boot to support 10K+ concurrent users with 99.9% uptime.
- Designed RESTful APIs for cross-service communication, reducing latency by 30% during peak loads.
- Optimized PostgreSQL and MongoDB operations through indexing and query restructuring, improving data access performance by 40%.
- Containerized applications using Docker and deployed via Jenkins CI/CD pipelines, reducing release overhead and improving reliability.
- Participated in on-call rotations and root cause analysis for production issues, reducing repeated outages by 25%.
- Implemented end-to-end monitoring and automated release validation using CloudWatch and Jenkins, improving system reliability and reducing incident response time by 30%.

Software Developer Intern | *Wipro, India*

Feb 2022 - Sep 2022

- Designed and built microservices using Spring Boot for an enterprise e-commerce platform, enabling independent feature deployment and faster release cycles.
- Improved backend efficiency by refactoring Java-Hibernate data flows and tuning queries, reducing order execution time by 25%.
- Created JUnit and Mockito-based test suites achieving 90% coverage, preventing regressions in CI/CD pipelines.
- Integrated third-party payment and inventory services through secure REST APIs, improving transaction reliability and reducing timeout failures by 20%.
- Collaborated with DevOps to containerize services using Docker and automate deployments via Jenkins, cutting manual release effort by 40%.

PROJECTS

Distributed Task Scheduler Using gRPC | *C++, gRPC, Protobuf, Kafka, Kubernetes, CMake*

- Built a low-latency distributed task scheduler achieving less than 5 ms dispatch time for 10 K+ concurrent tasks.
- Implemented dynamic worker reassignment, real-time failure detection, and shared-memory coordination using POSIX locks.
- Deployed on Kubernetes for fault-tolerant orchestration and elastic scaling.

Fine-Tuning Large Language Models (LLMs) | *Python, Lambda Labs, FastAPI, Docker, Hugging Face*

- Deployed LLaMA 2 on Lambda Labs for reproducible inference with sub-5 ms latency.
- Applied quantization and pruning to shrink model size by 70 % while preserving accuracy.
- Enhanced inference throughput by 50 % through distillation and neural architecture search; containerized with Docker.

Movie Ticket Booking Application | *React.js, Redux, Node.js, AWS, Redis, MongoDB, TypeScript*

- Designed an event-driven microservice architecture scaled to 1 K+ concurrent users.
- Implemented RabbitMQ handling 5 K msgs/min, designed GraphQL schemas reducing over-fetching by 40 %.
- Used Redis distributed locks to prevent over-booking and ensure data consistency.

TaskFlow – Scalable Task Management API | *Java, Spring Boot, PostgreSQL, AWS EC2, Jenkins, GitHub Actions*

- Developed a RESTful API backend enabling multi-user collaboration and role-based access control.
- Deployed on AWS EC2 with CI/CD pipelines (GitHub Actions + Jenkins) achieving zero-downtime releases.

Serverless Image Analyzer (AI + AWS Lambda) | *Python, AWS Lambda, S3, API Gateway, CloudWatch*

- Built a serverless image-classification service using pre-trained CNN models; processed images in less than 500 ms.
- Integrated S3 for storage and API Gateway for endpoints; achieved 99 % classification accuracy.
- Implemented CloudWatch logging and alerting for observability and fault analysis.

CERTIFICATIONS & PUBLICATIONS

- *"Voice-Controlled Robotic Vehicle"* – G. Ashritha et al., *International Journal of Computer Science Trends and Technology (IJCTST)*, Vol. 11, Issue 2, 2023. Link: ijctstjournal.org
- **Python Programming (Cisco, Internshala)** – Trained in OOP, data structures, scripting, and automation.

EDUCATION

Master of Science in Information Technology | *University of Massachusetts, Boston, MA*

Sep 2023 - May 2025

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Design, Distributed Systems, Cloud Computing, Database Management Systems, Operating Systems, Microservices Architecture, Advanced Java Programming, Machine Learning, Data Engineering, Software Design Patterns, Web Application Development, Parallel & Concurrent Programming.